MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, POWER, ANALOG, KEY CODE HEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic power amplifier module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 9111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.163 failures per million operating hours for class I or 0.507 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HEE.

3.5.2 Size. The module size shall be 1B, format A.

3.5.3 Weight. The module shall have a maximum weight of 95 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned 1A1 through 1A20. Contact pins 1A21 through 1A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/219-1 for class I and M28787/219-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

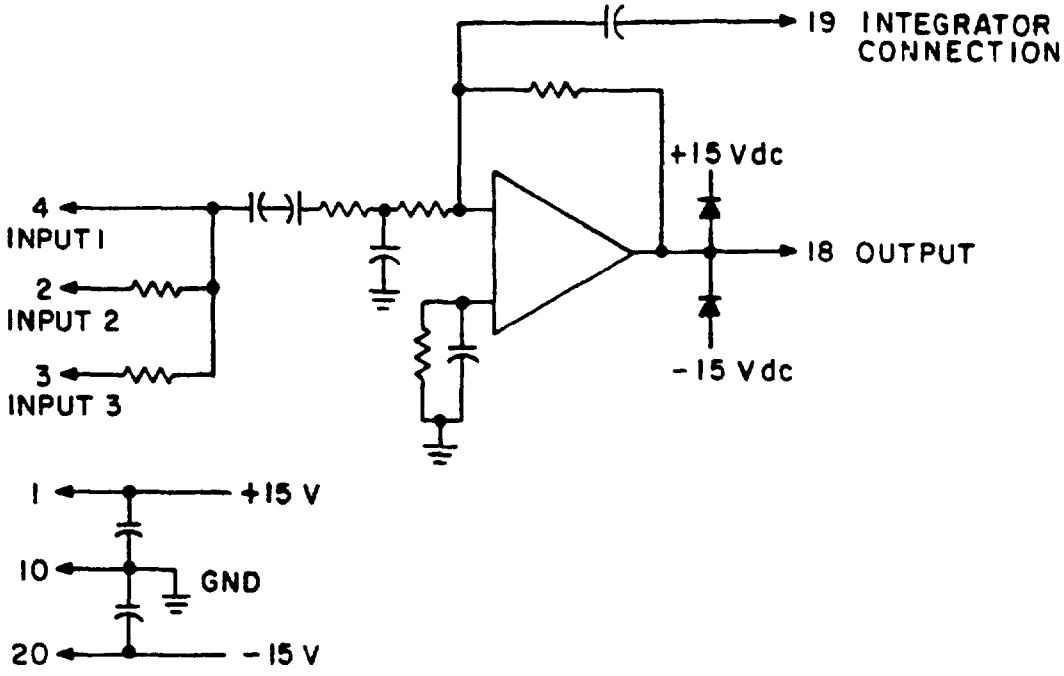
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE: The following pins are connected to pin 10 (ground): 5-9, 11-17.

FIGURE 1. HEE module function diagram, power amplifier.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	Stress	III	Q	10	15	---	---	---	---	---	---	---	---	s
C1	Output current limit	III	S	2 4	4 8	2.4	4.8	2.4	4.8	2 4	4.8	2.4	4.8	V pk
C2	Output current limit	III	S	-4 8	-2.4	-4.8	-2.4	-4.8	-2.4	-4 8	-2.4	-4.8	-2.4	V pk
D1, D2	Quiescent power supply current	IV	X	0 5	35.0	0 5	35.0	0.5	35.0	0 5	35.0	0.5	35.0	mA
E1	Gain (Gain = 5)													
	Initial	V	X	0 165973	0.167362	0.165809	0.167532	0.165700	0.167644	0.165425	0.167926	0.165508	0.167841	Ratio
	End of life	V	X	0 165617	0 167728	0 165453	0.167897	0 165234	0.168123	0 165070	0.168293	0.165070	0.168293	Ratio
E2	Gain (Gain = 6.67)													
	Initial	V	X	0.129812	0 130946	0.129678	0 131084	0.129589	0.131176	0.127365	0.131406	0.129432	0 131337	Ratio
	End-of-life	V	X	0.129522	0 131245	0.129387	0.131383	0 129209	0.131568	0 129076	0.131706	0.129076	0.131706	Ratio
E3	Gain (Gain = 10)													
	Initial	V	X	0 090496	0 091323	0 090399	0.091424	0 090334	0.091491	0.090171	0.091659	0.090220	0.091608	Ratio
	End of-life	V	X	0 090285	0 091541	0.090187	0.091642	0.090057	0.091776	0.089960	0.091877	0.089960	0 091877	Ratio
E4	Gain, integrater													
	Initial	V	X	0.090541	0 091393	0 090496	0.091439	0 090496	0.091439	0.090454	0.091485	0.090481	0.091460	Ratio
	End of-life	V	X	0.090372	0.091550	0.090291	0.091649	0.090291	0.091649	0 090250	0.091693	0.090250	0.091693	Ratio
F1 F8, F11, F12	Frequency response	VI	X	0.97	1 05	0.97	1.05	0.97	1.05	0.97	1.05	0.97	1 05	V rms
F9, F10	Frequency response	VI	X	0 70	1 05	0.70	1.05	0.70	1.05	0.70	1.05	0.70	1.05	V rms
F13	Frequency response	VI	X	0.95	1 05	0.95	1.05	0 95	1.05	0 95	1 05	0.95	1.05	V rms
F14, F15	Frequency response	VI	X	0 80	1 05	0.80	1.05	0.80	1.05	0.80	1.05	0.80	1 05	V rms

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
F16	Frequency response	VI	X	0.69	1.05	0.69	1.05	0.69	1.05	0.69	1.05	0.69	1.05	V rms
F17	Frequency response	VI	X	0.67	1.05	0.67	1.05	0.67	1.05	0.67	1.05	0.67	1.05	V rms
F18	Frequency response	VI	X	0.60	1.05	0.60	1.05	0.60	1.05	0.60	1.05	0.60	1.05	V rms
F19	Frequency response	VI	X	0.50	1.05	0.50	1.05	0.50	1.05	0.50	1.05	0.50	1.05	V rms
F20	Frequency response	VI	X	0.03	1.05	0.03	1.05	0.03	1.05	0.03	1.05	0.03	1.05	V rms
G	Noise	VII	Q	0	100	0	100	0	100	0	100	0	100	mV pk
H1	Phase accuracy	VIII	X	181	186	181	186	181	186	181	186	181	186	Degree
H2	Phase accuracy	VIII	X	178	183	178	183	178	183	178	183	178	183	Degree
J	Output voltage swing	IX	X	5.6	9.7	5.6	9.7	5.6	9.7	5.6	9.7	5.6	9.7	V rms
K	Oscillation	IX	X	←		There shall be no oscillation present -							→	---
L	Input impedance check	X	S	2.48	2.52	2.48	2.52	2.48	2.52	2.48	2.52	2.48	2.52	V rms
M	Output offset voltage	XI	X	-13	13	-13	13	-13	13	-13	13	-13	13	mV
N1, N2	Power supply capacitance	XII	X	5.4	8.2	5.4	8.2	5.4	8.2	5.4	8.2	5.4	8.2	μF
P1, P2	Power supply ripple rejection	XIII	Q	0	10	0	10	0	10	0	10	0	10	mV
Q1, Q2	Diode forward voltage	XIV	S	0.4	1.0	0.4	1.0	0.4	1.0	0.4	1.0	0.4	1.0	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence. If test conditions A, B, C are performed during quality assurance testing they shall be followed by performance of at least all 100 percent tests.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through F and H through N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through F and H through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A and C through Q shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Part number M28787/219-1 supersedes part number M28787/219. The two parts are interchangeable.

6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation	1.8	1.8	W
Power supply voltage at 35 mA maximum	±13.5 min, ±16.5 max	±13.5 min, ±16.5 max	V
Output voltage, R_L = 100Ω	5.6 min	5.6 min	V rms
Output current	800 min, 1600 max	800 min, 1600 max	mA
Bandwidth			
Nonintegrating	>15	>15	kHz
Intergrating	30 to 1590	30 to 1590	Hz

6.6 Module function. This module contains an AC coupled power amplifier which can be pin programmed for three different gains. An optional integrator capacitor may be connected to the output terminal. The gain options are 5, 6.67, and 10.

6.7 Caution. This module may be subject to hot socket damage. Remove power before inserting this module.

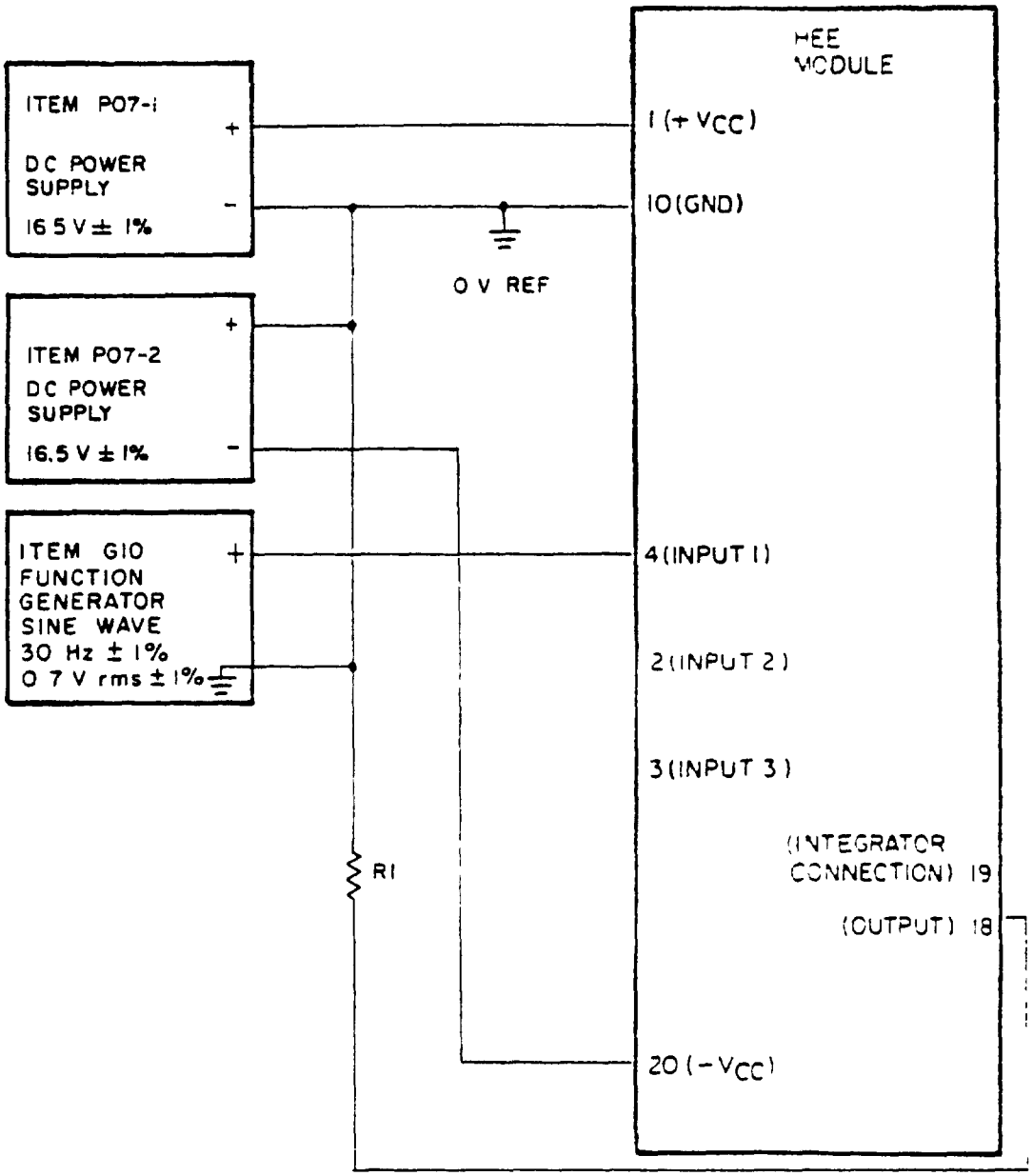
6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTE R1 = 100Ω ±1 , 10 W.

FIGURE 2 Operating temperature, component temperature, and life test circuit

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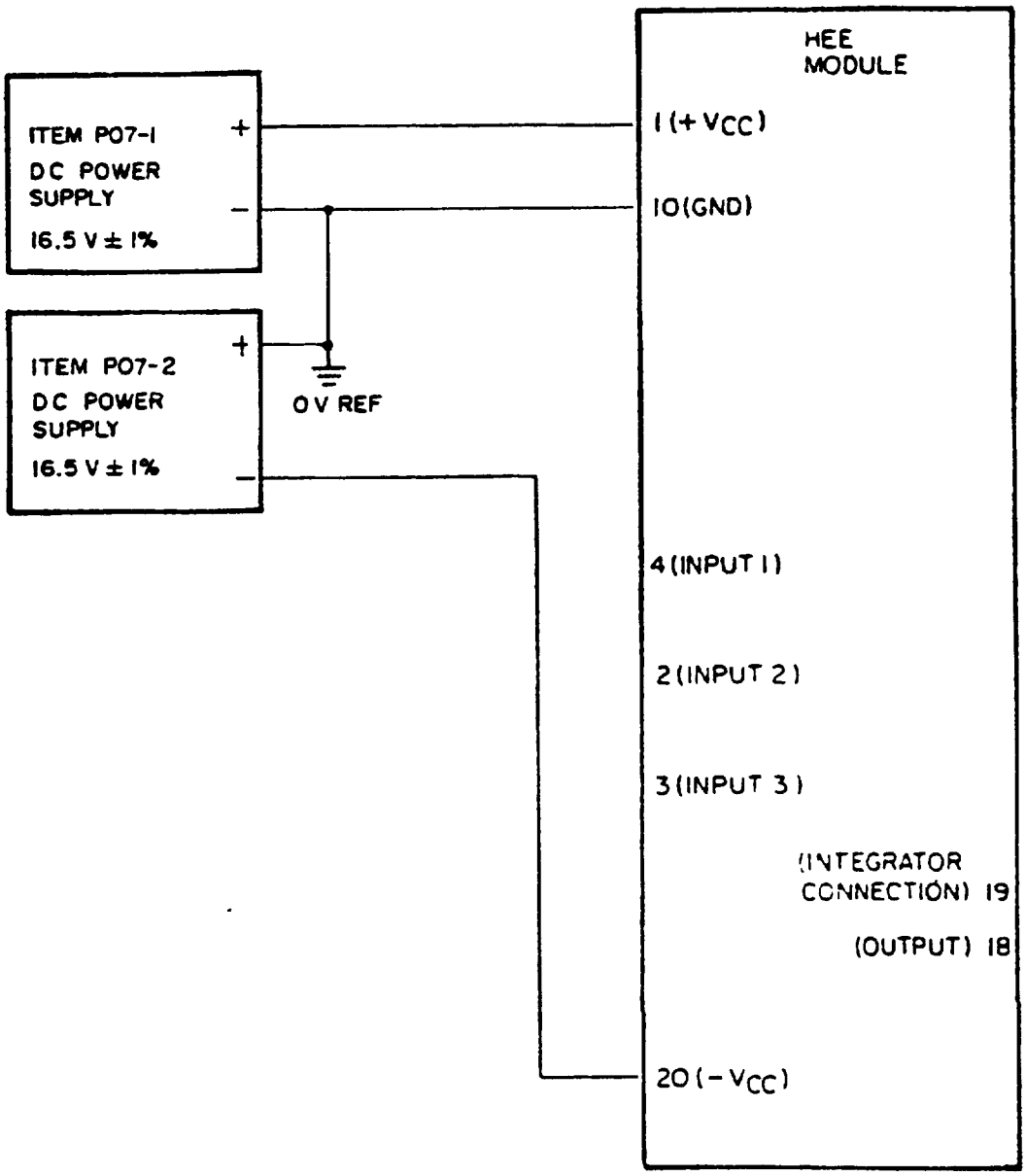


FIGURE 3. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation, frame	All pins	Frame	10	∞	MΩ	4

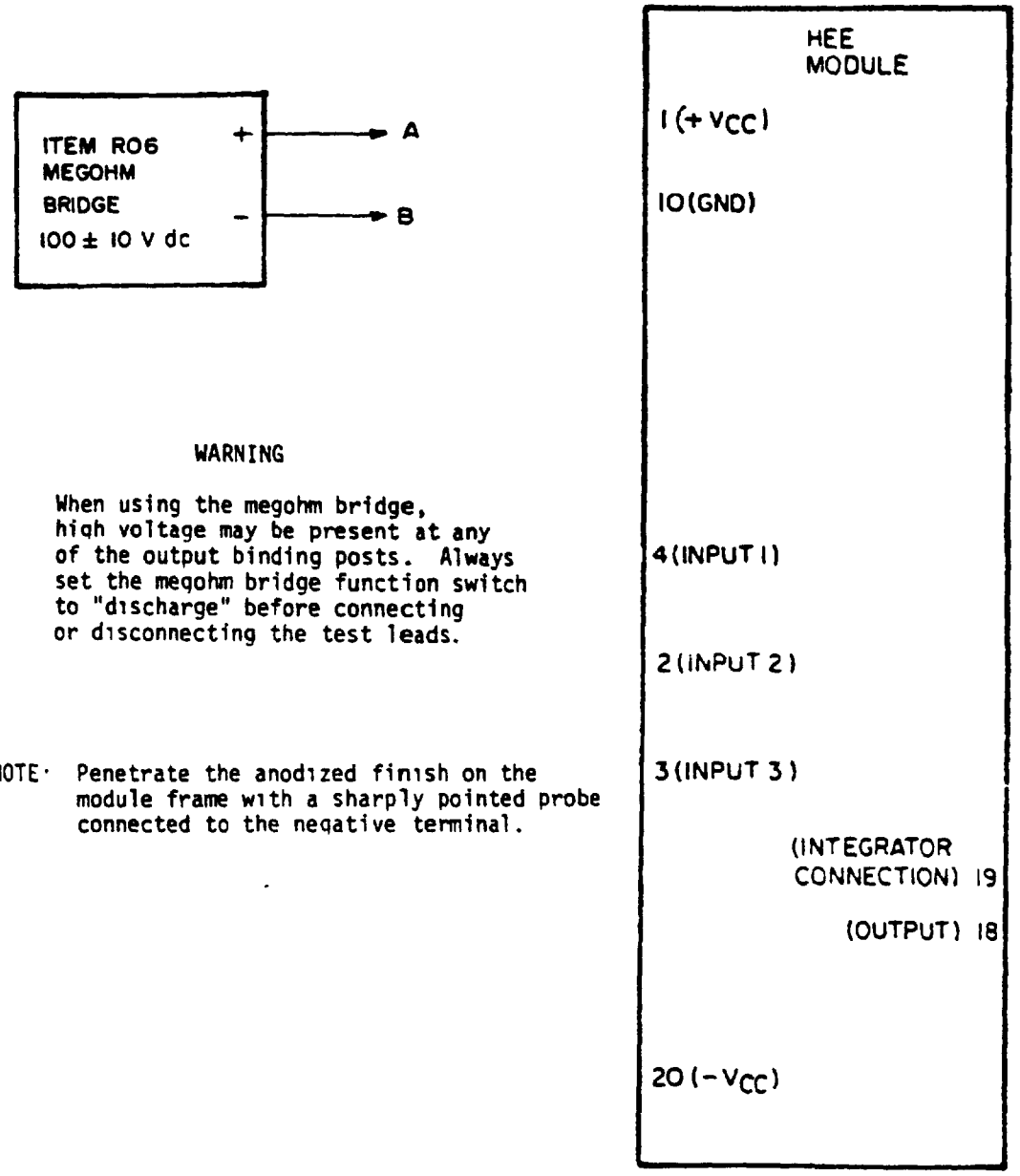
TABLE III Stress and output current limit test

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B	Stress	18	10	15	s	5
C1	Output current limit (+ peak)	Open	2.4	4.8	V pk	5, 6
C2	Output current limit (- peak)	Open	-4.8	-2.4	V pk	5, 6

TABLE IV Quiescent power supply current test

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D1	Quiescent power supply current	0.5	35.0	mA	/
D2	Quiescent power supply current	0.5	35.0	mA	/

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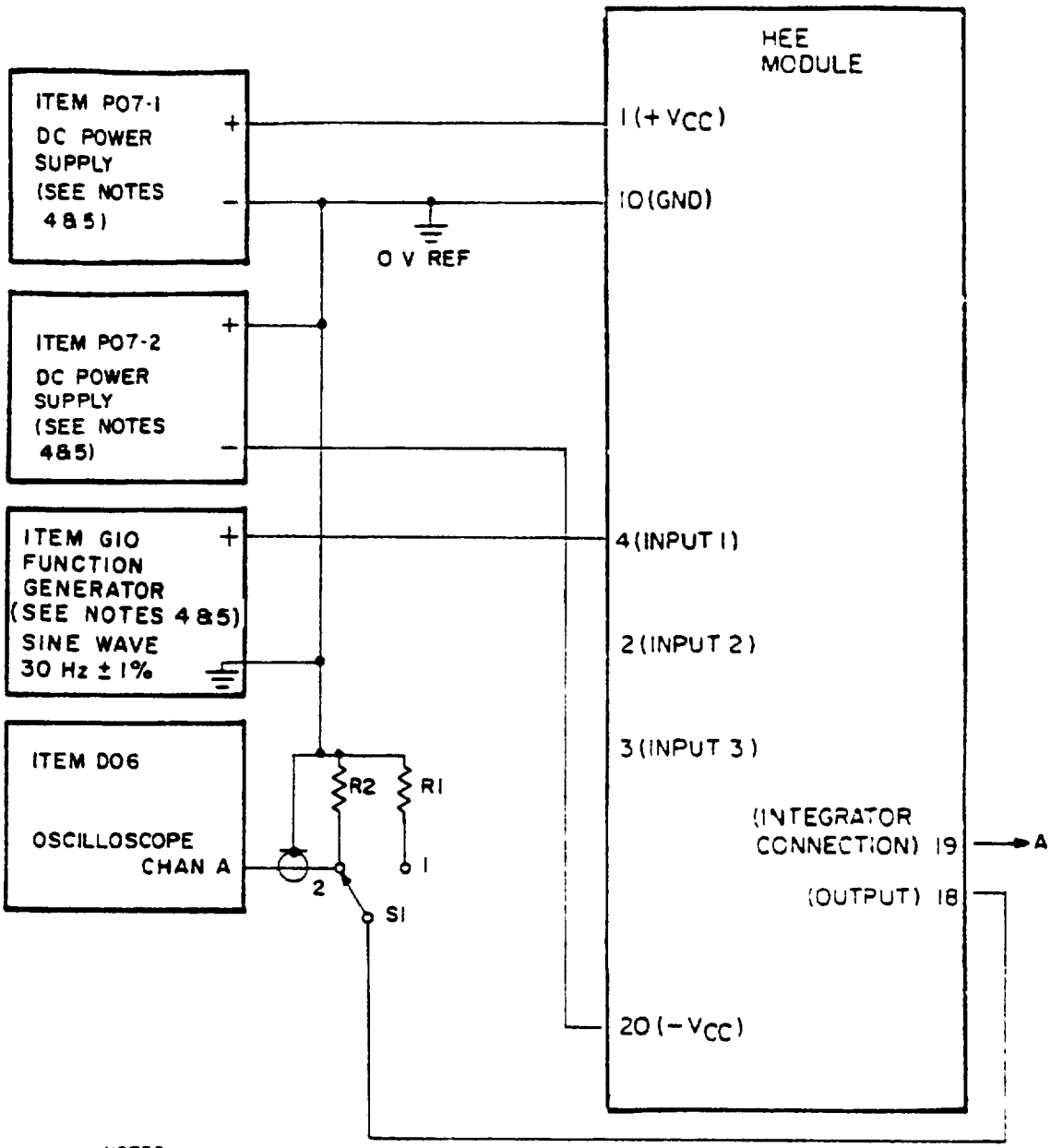
WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 4. Isolation test circuit for table II.

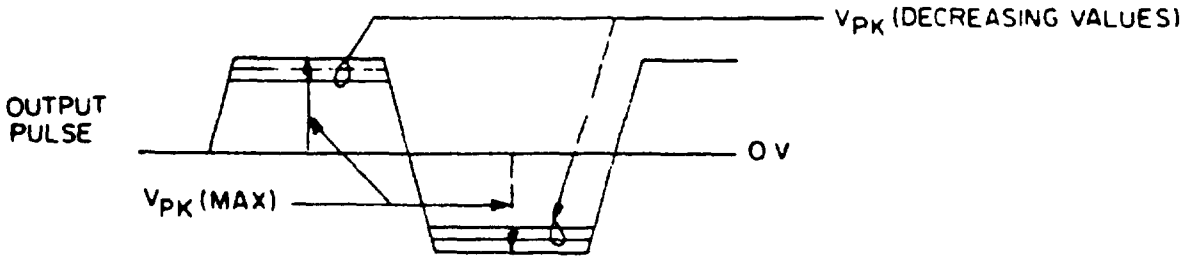
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NOTES.
1. R1 = 10^ ±1., 10 W.
2. R2 = 3^ ±1 , 5 W.
3. S1 = SPDT switch.
4. Test condition B:
   a. Switch S1 in position 1.
   b. Items P07-1 and P07-2 set to 17.8 V ±1%.
   c. Item G10 output voltage set to 0.70 ±1% V rms.
   d. The module shall be subjected to stress for a minimum of 10 seconds and a maximum of 15 seconds.
5. Test condition C
   a. Switch S1 in position 2.
   b. Items P07-1 and P07-2 set to 16.5 V ±1%.
   c. Item G10 output voltage set to 0.53 ±1% V rms.
   d. The module shall not be operated for more than 10 seconds.
   e. The table III limits shall be the maximum value indicated on item D06 as shown on figure 6.

FIGURE 5 Stress and output current limit test circuit for table III.

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NOTE: Since output amplitude decreases due to resistive heating, the measurements shall be taken during the first cycle of the output.

FIGURE 6. Output current limit test circuit waveform for test conditions C1 and C2 for table III.

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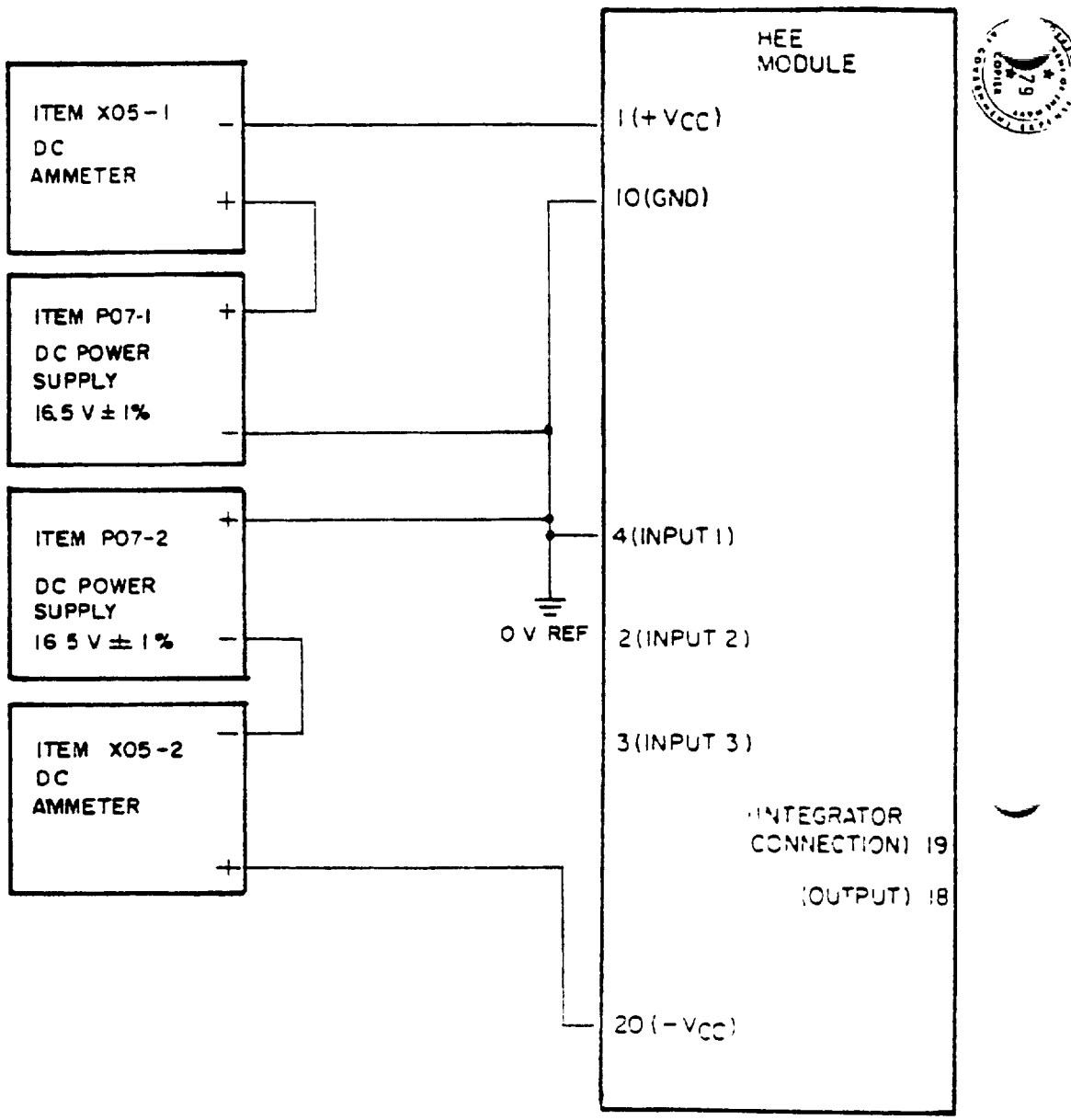


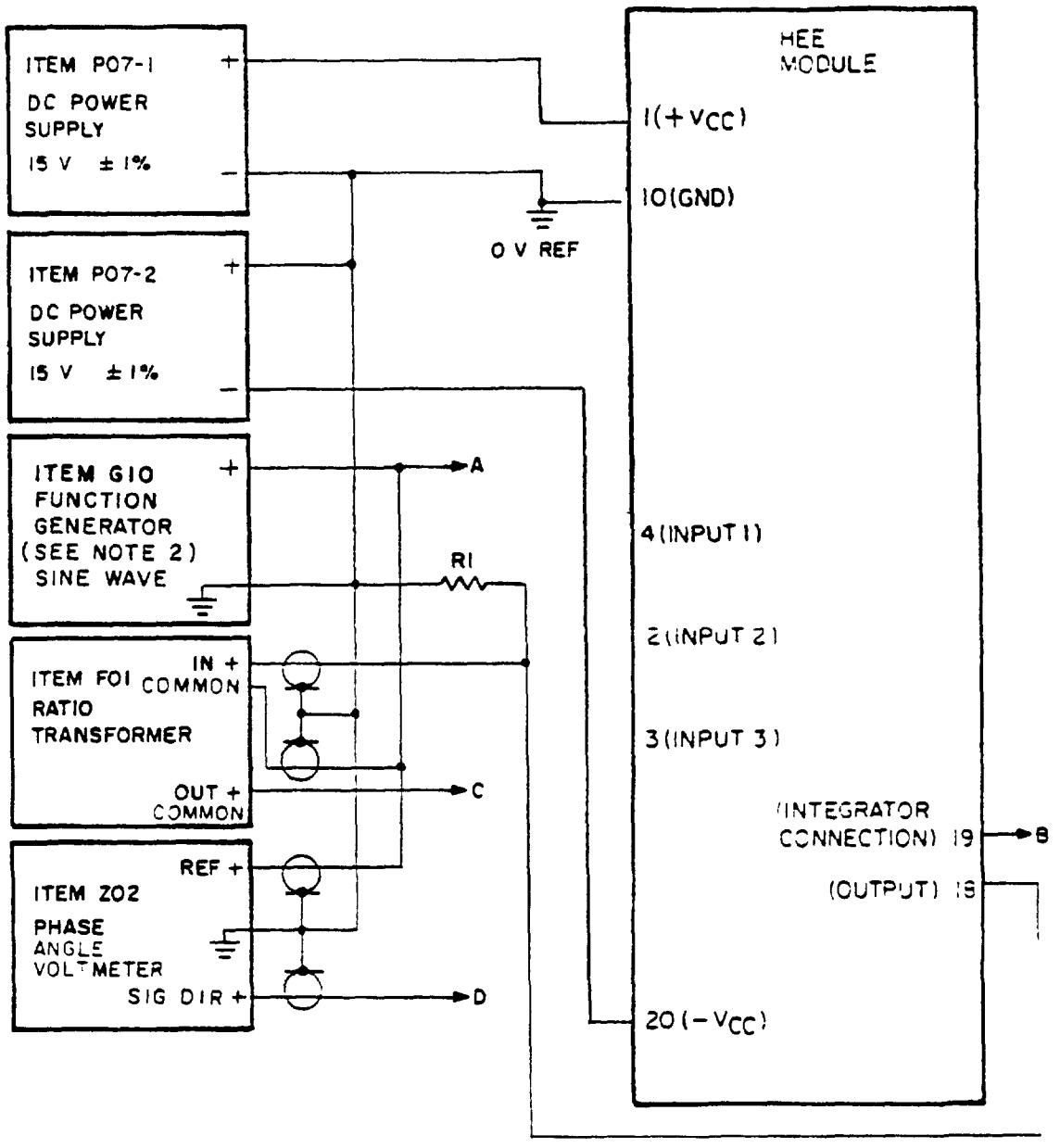
FIGURE 7 Quiescent power supply current test circuit for table IV

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TABLE V. Gain test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Gain (Gain = 5)	3	Open	.165973	.167362	Ratio	8
E2	Gain (Gain = 6.67)	2	"	.129812	.130946	"	"
E3	Gain (Gain = 10)	4	"	.090496	.091323	"	"
E4	Gain, integrater	4	18	.090541	.091393	"	"

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NOTES

1. RI = 100 ±1 , 1 W.
2. Set item G10 at 1 V rms ±1 at 400 Hz ±1, for test conditions E1 and E2 and 0.5 V rms ±1 at 400 Hz for test conditions E3 and E4 first then perform the remainder of the test as specified in each test condition.
3. Adjust the reference on item Z02. Some instruments may not reach the reference mark. In that case, adjust the reference for the maximum possible.
4. Set the function switch on item Z02 to 0° and connect point D to pin 18. Adjust the phase dial for a null on item Z02.
5. Remove point D from pin 18 and connect point D to point C. Set the function switch to 90° and adjust item F01 for a null on item Z02. The reading on item F01 for each test condition shall meet the limits of table V.
6. The output common terminal of item F01 shall not be connected at any time
7. Gain = $\frac{1 - (\text{table V limits})}{(\text{table V limits})}$ for test conditions E1-E3.
8. For test condition E4 Gain = $\frac{\cos\theta}{(\text{table V limit} - \cos\theta)}$ where $\theta = 180° -$ phase angle of output.

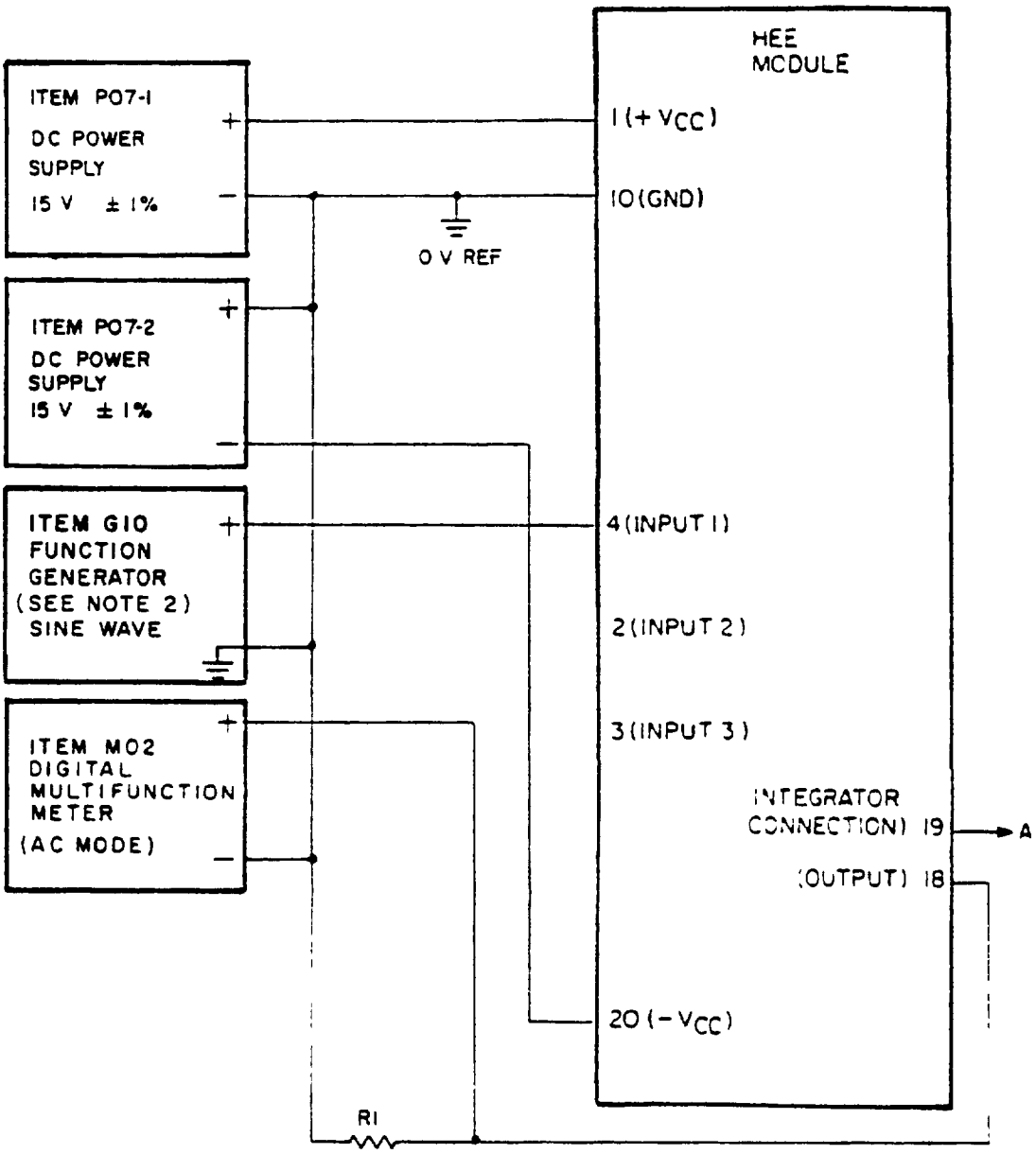
FIGURE 3 Gain test circuit for table V

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TABLE VI. Frequency response test.

Test condition	Parameter	Item G10 frequency Hz ±1%	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F1	Frequency response	50	Open	0.97	1.05	V rms	9
F2	"	60	"	"	"	"	"
F3	"	100	"	"	"	"	"
F4	"	400	"	"	"	"	"
F5	"	800	"	"	"	"	"
F6	"	1000	"	"	"	"	"
F7	"	1500	"	"	"	"	"
F8	"	2000	"	"	"	"	"
F9	"	15000	"	0.70	"	"	"
F10	"	50000	"	0.70	"	"	"
F11	"	50	18	0.97	"	"	"
F12	"	60	"	0.97	"	"	"
F13	"	400	"	0.95	"	"	"
F14	"	800	"	0.80	"	"	"
F15	"	1000	"	0.80	"	"	"
F16	"	1590	"	0.69	"	"	"
F17	"	1700	"	0.67	"	"	"
F18	"	2000	"	0.60	"	"	"
F19	"	2500	"	0.50	"	"	"
F20	"	50000	"	0.03	"	"	"

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NOTES.
1. R1 = 100Ω ±1%, 1 W.
2. Set item G10 output voltage to 0.10 V ±1% and frequency as specified in table VI.

FIGURE 9. Frequency response test circuit for table VI.

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TABLE VII. Noise test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
G	Noise	0	100	mV pk	10

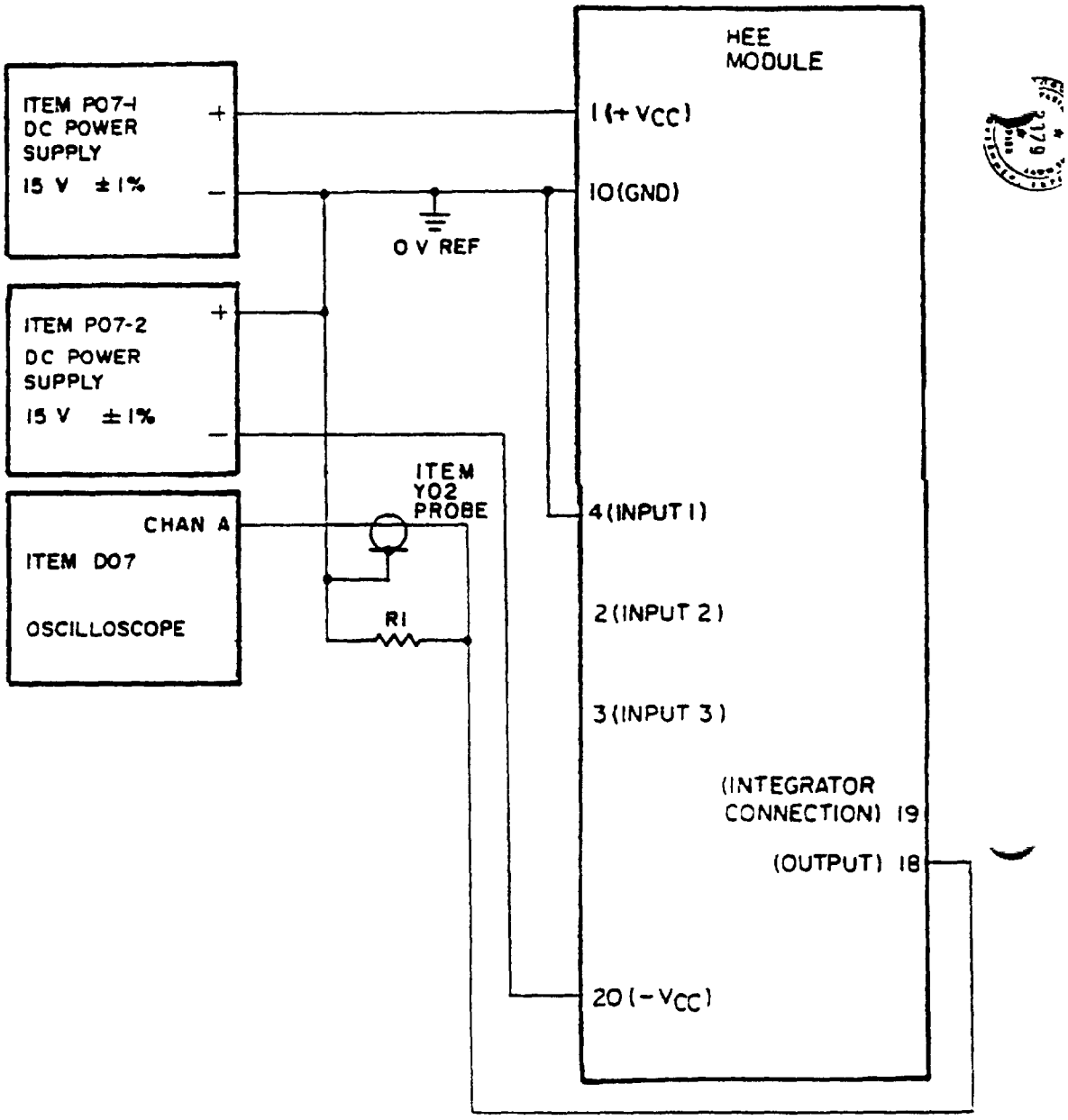
TABLE VIII. Phase accuracy test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
H1	Phase accuracy (phase angle at 30 Hz)	181	186	Degree	11
H2	Phase accuracy (phase angle at 120 Hz)	178	183	Degree	11

TABLE IX. Output voltage swing and oscillation test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J	Output voltage swing (positive)	Open	5.6	9.7	V rms	12
K	Oscillation	18, 19	See figure 12, note 5		V rms	12

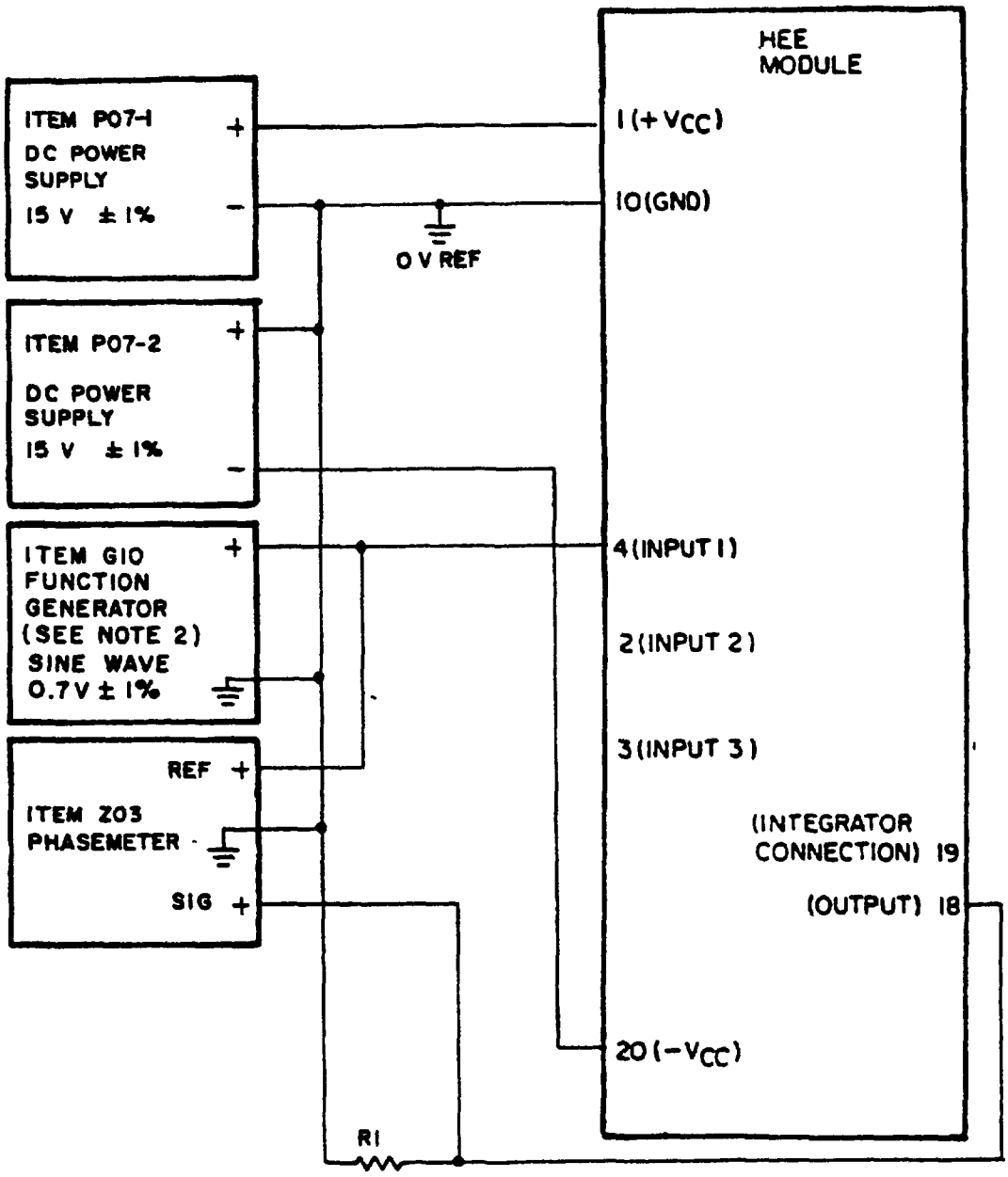
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NOTES.
1. R1 = 100Ω ±1%, 1 W.
2. Set the item D07 time base to 1 second per centimeter and observe the output peaks both above and below the output dc level for 10 seconds minimum.

FIGURE 10 Noise test circuit for table VII.

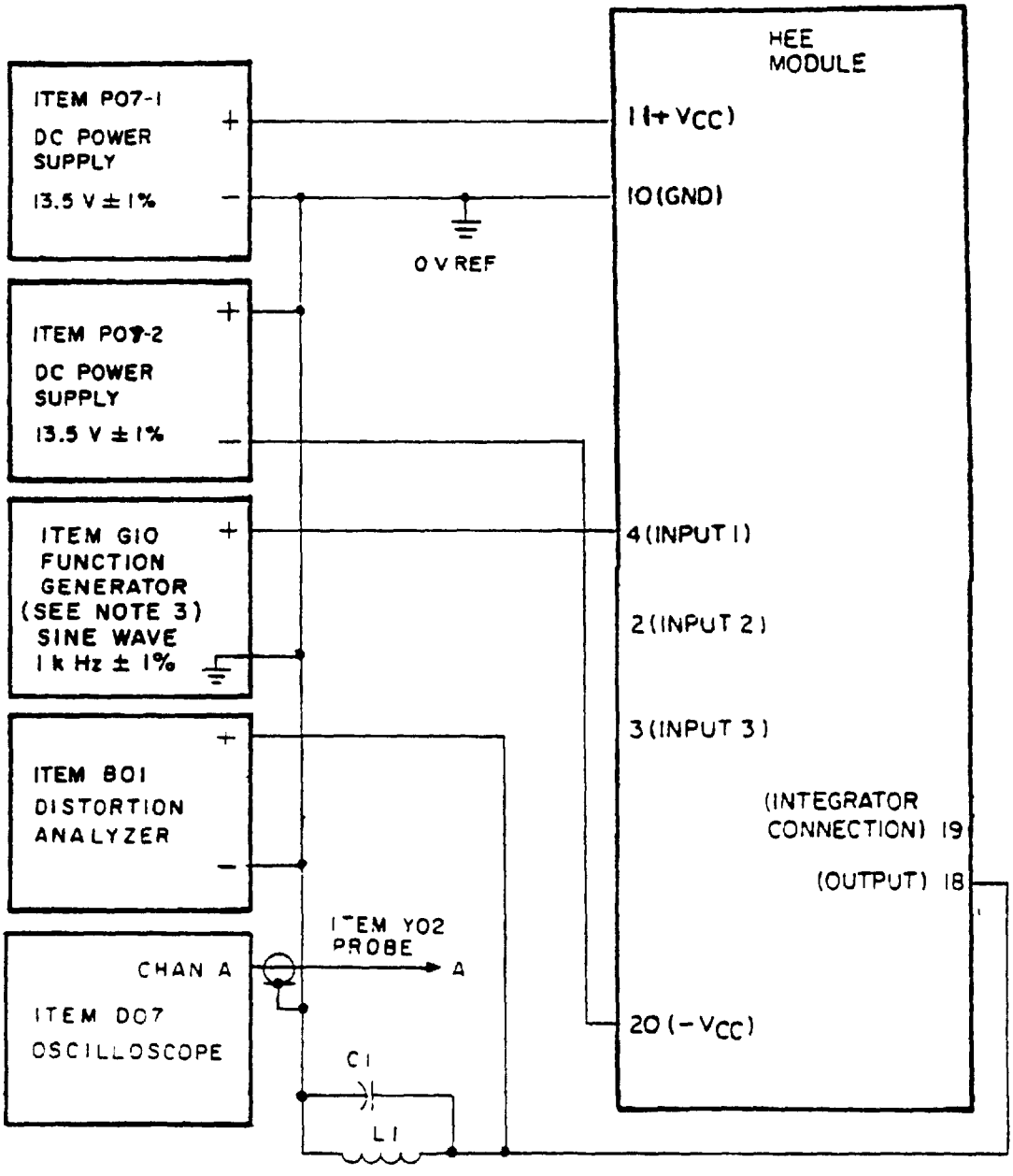
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NOTES:
1. R1 = 100Ω ±1%, 1 W.
2. Set item G10 to 30 Hz ±1% for test condition H1 and to 120 Hz ±1% for test condition H2.
3. The output phase angle shall be positive (lead).

FIGURE 11. Phase accuracy test circuit for table VIII.

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NOTES:
1. C1 = 0.047 µF ±10%.
2. L1 = 39 mH ±10%.
3. Adjust item G10 amplitude until item B01 reads 5.0 ±0.5 V rms. Readjust the level on item B01 and again adjust item G10 until item B01 reads 1%. Switch item B01 to voltmeter and read the voltage. Readjust the reference level on item B01 as required. The voltage shall meet the limits of table IX.
4. This test also verifies distortion to be less than 1%.
5. Connect item D07 (point A) as specified in table IX. There shall be no oscillation present on the waveform.

FIGURE 12. Output voltage swing and oscillation test circuit for table IX.

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TABLE X. Input impedance check test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Input impedance check	2.48	2.52	V rms	13

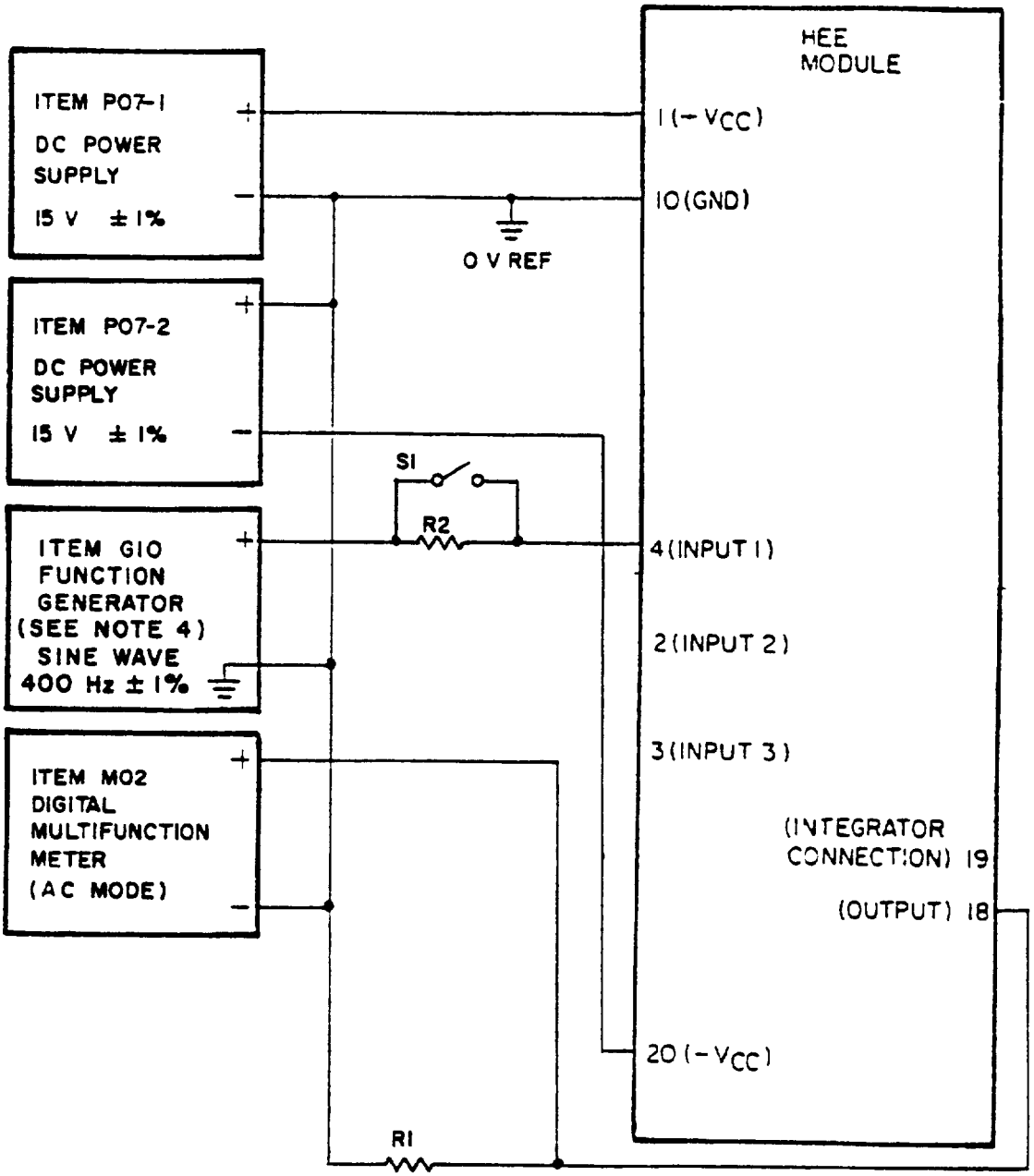
TABLE XI. Output offset voltage test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Output offset voltage	-13	13	mV	14

TABLE XII. Power supply capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Power supply capacitance ($+V_{CC}$)	1	10	5.4	8.2	µF	15
N2	Power supply capacitance ($-V_{CC}$)	10	20	5.4	8.2	µF	15

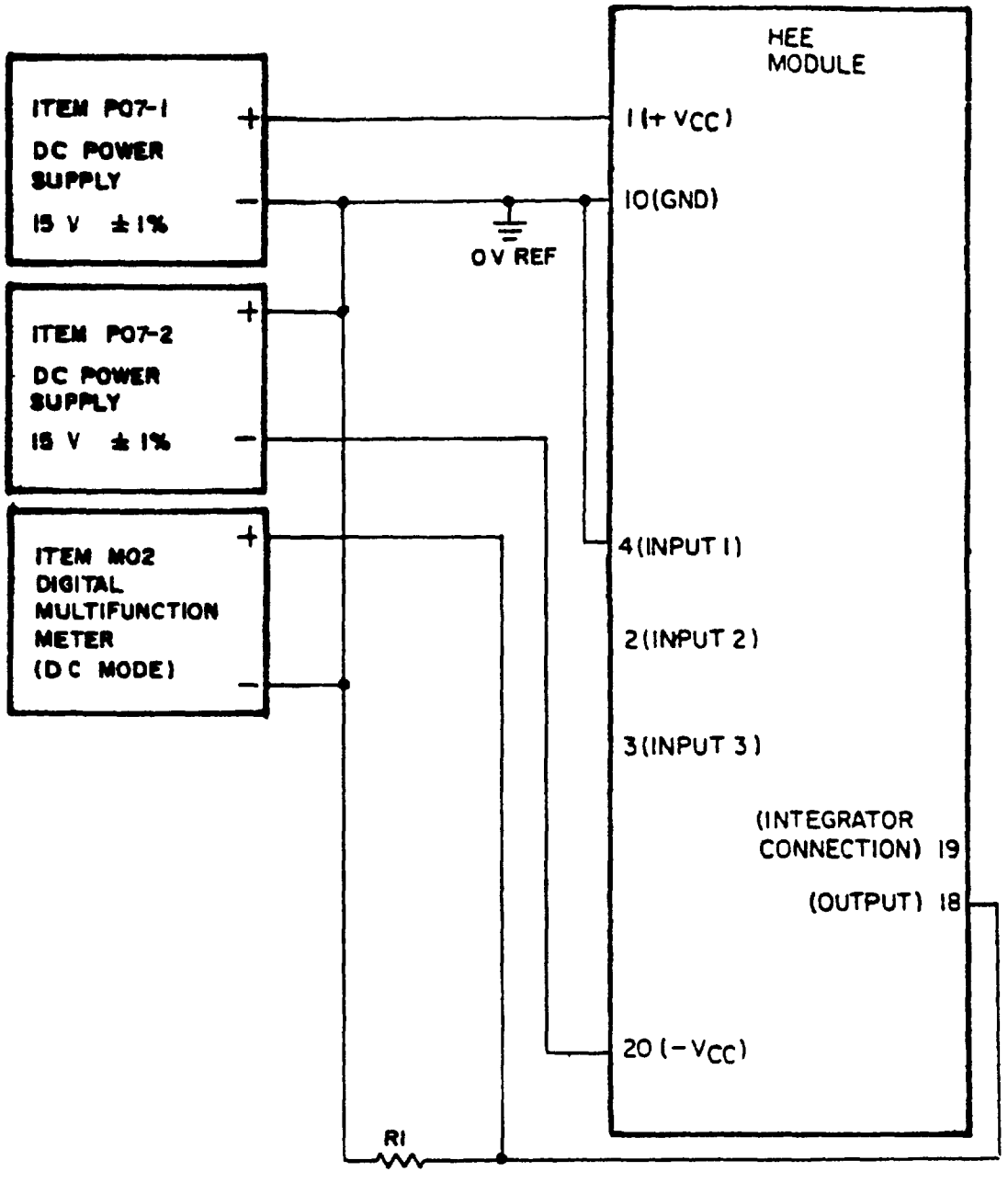
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NOTES:
1. R1 = 100Ω ±1%, 1 W.
2. R2 = 10 kΩ ±1%, 1/8 W.
3. S1 = SPST switch.
4. Adjust item G10 to obtain 5.00 ±0.05 V rms output voltage at pin 18 with switch S1 closed. Then open switch S1. The output shall conform to the table X limits

FIGURE 13. Input impedance check test circuit for table X.

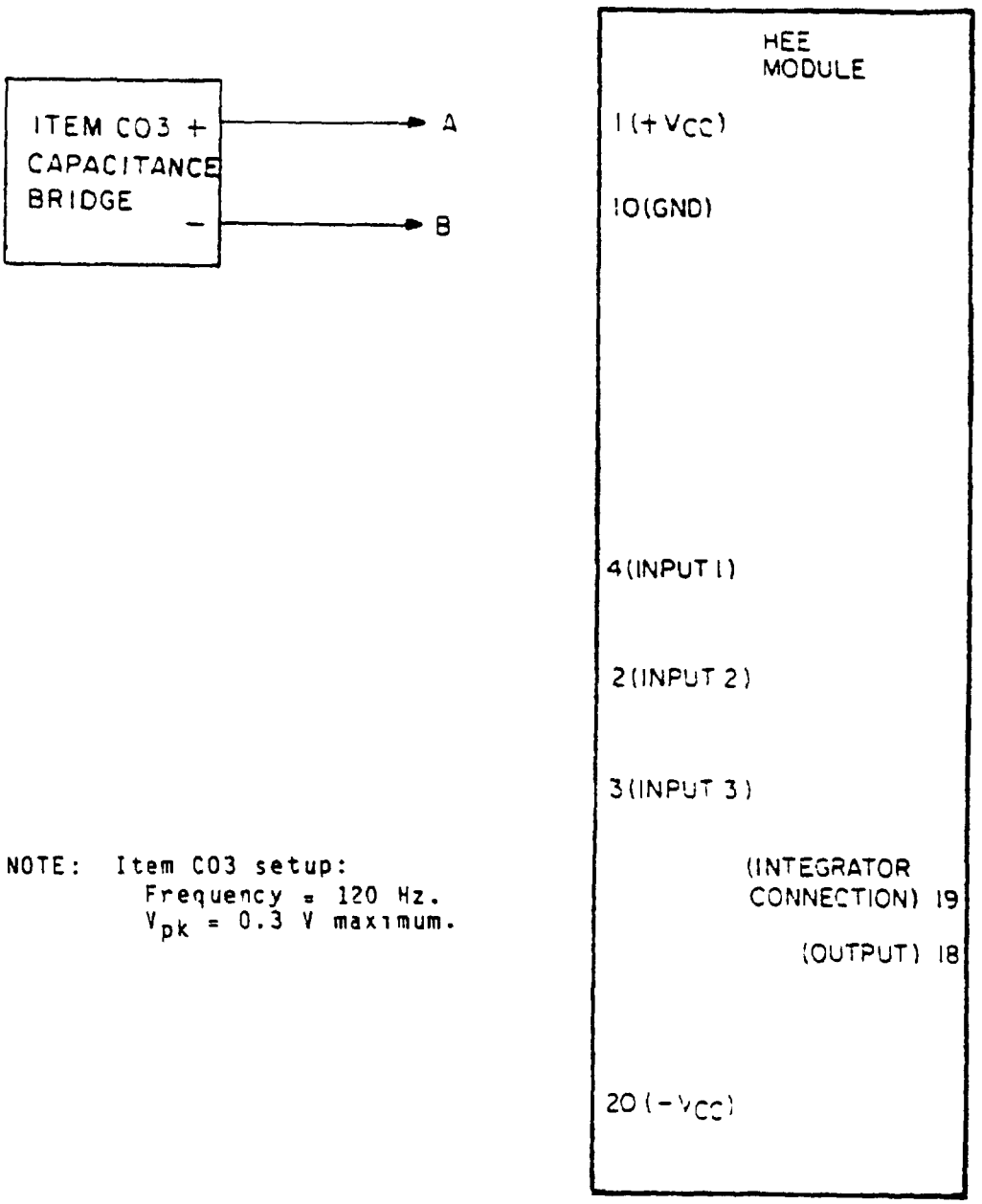
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NOTE: R1 = 100Ω ±1%, 1 W.

FIGURE 14. Output offset voltage test circuit for table XI .

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NOTE: Item CO3 setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 15. Power supply capacitance test circuit for table XII.

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TABLE XIII. Power supply ripple rejection test.

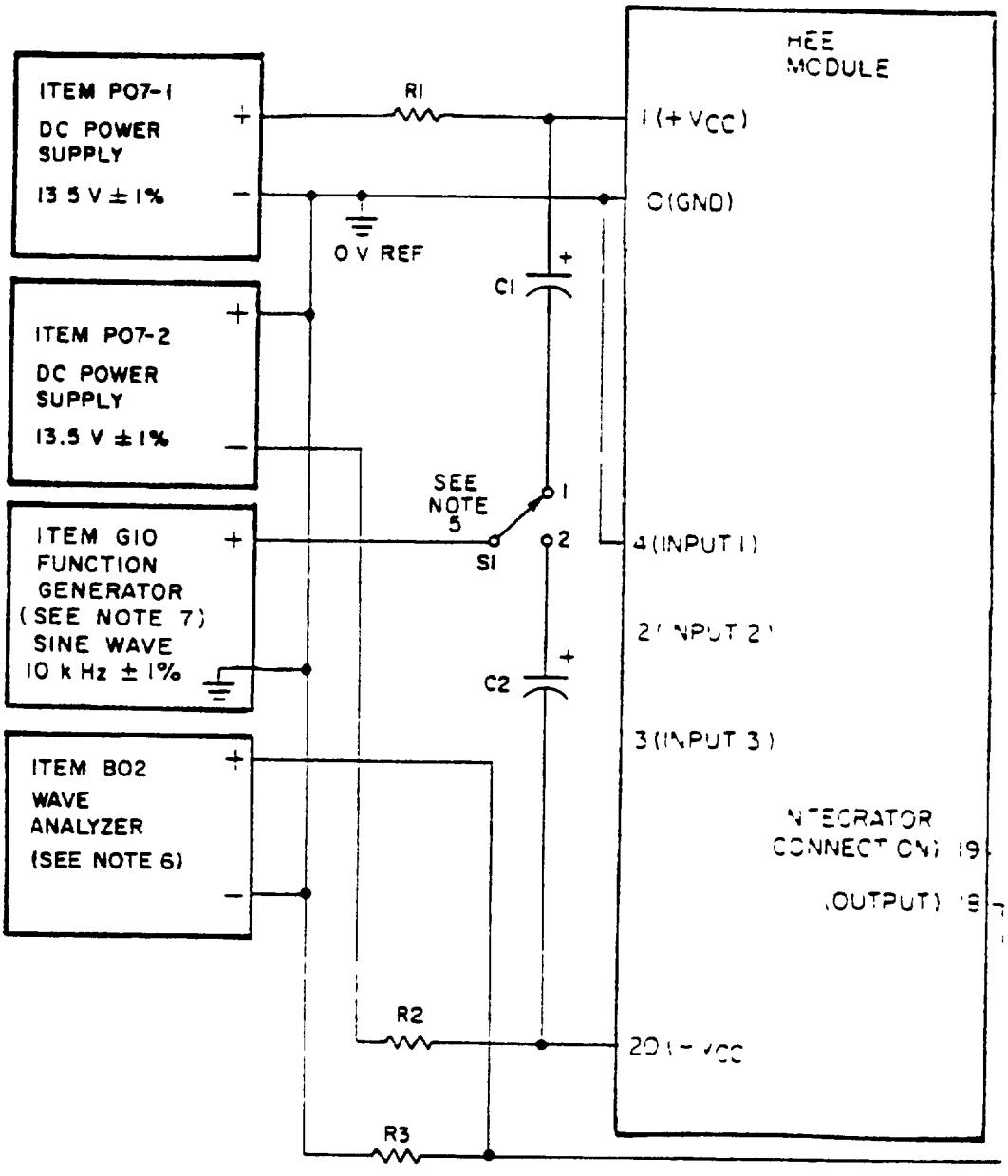
Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply ripple rejection ($+V_{CC}$)	0	10	mV	16
P2	Power supply ripple rejection ($-V_{CC}$)	0	10	mV	16

TABLE XIV. Diode forward voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	Diode forward voltage	18	1	0.4	1.0	V	17
Q2	Diode forward voltage	20	18	0.4	1.0	V	17

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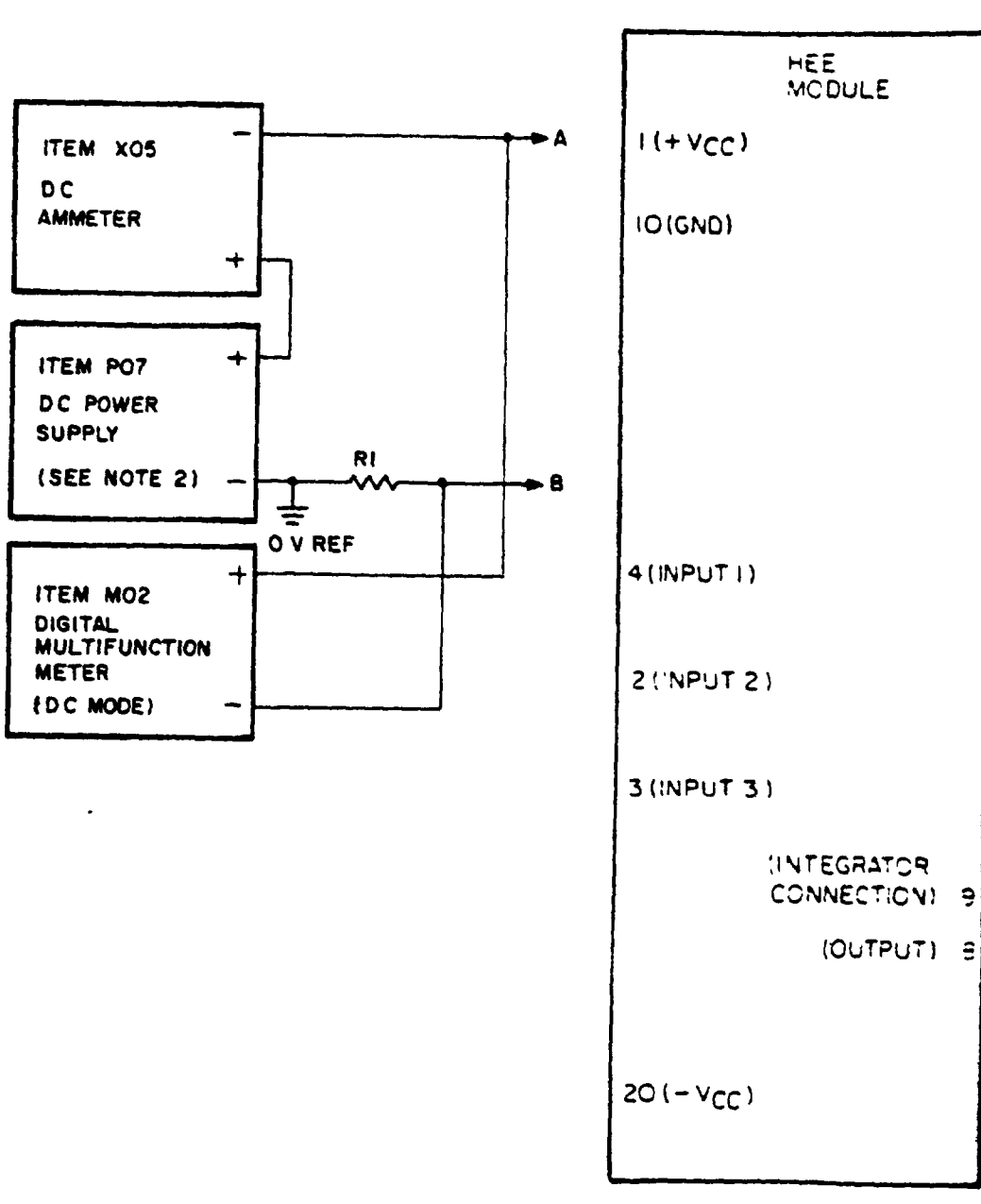
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NOTES.
1. R1 = R2 = 47 ±5 , 1/4 W.
2. R3 = 100 ±1 , 1 W.
3. C1 = C2 = 20 μF ±20 , 20 V dc.
4. S1 = SPDT switch.
5. Switch S1 to position 1 for test condition P1 and to position 2 for test condition P2.
6. Tune item B02 to the item G10 frequency.
7. Adjust item G10 to obtain 150 mV rms ±1 at pins 1 and 20 for test conditions P1 and P2, respectively.

FIGURE 16. Power supply ripple rejection test circuit for table XIII.

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NOTES:
1. RI = 50Ω ±5%, 1 W.
2. Adjust item P07 for a current of 100 mA ±1% and read the voltage on item M02.

FIGURE 17. Diode forward voltage test circuit for table XIV.

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